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说话人 1 00:00:00

Great.This, okay, sorry.Here's the Mark.The right is what the progress and our question.And so the shuangshuang has completed the 3D more version on the 21st and send it via email.And there is a three questions we want to discuss.One is the about the TCO line in light, could it be splitted into two layers, like one upper level and one lower level?

说话人 2 00:00:55

What are you thinking?Let me just have a look.Yeah, from our side, are you thinking of trying to doubles, double stack one above the other? Or is it an offset?

说话人 1 00:01:17

let me double check with 双双 first, 范工, 哈喽, 范工在吗?

说话人 3 00:01:27

哎, 李博士, 啊, 在了。

说话人 1 00:01:28

您这个进出口分上下两层, 它是具体是一个什么想法呀? 能展示一下吗?

说话人 3 00:01:37

我共享一下屏幕啊。

说话人 3 00:01:54

上次我看 case 给咱们展示的他那边的模型, TCU 管路他排成了一排, 你看我这这不是排成两排了吗? 他那个空间好像比较大, 他有空间把它排成一排。我这个我就考虑以后保温什么的, 就只最好是两排, 他他能明白我的意思吗? OK。就是他那个, 他那上下口他都在一排上放, 就那十, 就这不是10

个管嘛，都在一层了。然后我因为我这边空间有限嘛，我想把它弄成两层，进口在下边，出口在上边，就相当于一层是5个，

说话人 1 00:02:29

So this, could you see Shuangshuang's screen? This one is what we're talking about, the two layers. And so the in light, maybe from the lower layer and the upper layer is for the outline. And cuz we thought the space maybe limited and we found you were thinking that making it within one line and shuangshuang was thinking that if we could.

说话人 1 00:02:56

Making it in two lines. The, the, we can reserve some space and also there are some benefits fit for the keeping warm function, the keeping the temperature. Yeah. Okay. And mainly consideration is about the space. Yeah, I'll believe so. Let me share our screen. Okay, sure.

说话人 2 00:03:34

No, it's not that. This similar, Peter, we were coming through in a line, but then as we come through, we in effect double stack it.

说话人 2 00:04:06

Can Shuang deal with a Navisworks file, an NWD?

说话人 1 00:04:16

范工，您这边能编辑 NWD 文件吗？叫 Navisworks 的文件类型。

说话人 3 00:04:26

NWD 它这也是一个三维吗？

说话人 1 00:04:29

对，应该是。我看看我们这个。我以前见过。

说话人 3 00:04:33

我我我等我打开这个，我看看我们这边有没有，能不能存成这个。或者说，case 把他这个现场的模型给我，我看我能不能和他对上也行。

说话人 1 00:04:44

行，OK。

说话人 3 00:04:46

我先看看我能不能打。

说话人 1 00:04:47

I think, I think NWD, could you send us a sample file like this file if it's in the NWD style and so that shuangshuang will try to open it and edit it. Okay.

说话人 2 00:05:07

From this is what we use. So when we insert the step file, we just insert. Your step file, and we've actually got some, we've connected up our pipes to your pipes on the original model. So this is quite an early Navisworks file. This location is where it comes through the wall. But obviously, it can come through a bit lower as all we do. If you're coming through, you might come straight into these pipes if that works for you height wise. But that was the area we had penciled in for the transition for the wall. So yeah, Swan can get some dimensions off of that. Then you could LA, you could do your wall plate based on that sort of area on the wall. Positions for this, but he hasn't come back with a drawing yet. But if you can get it off the Navisworks file, then it could be there. Does that make sense? Okay. Yeah, thanks. Let me rest of this cuz it's changed. It's just that, look, if you just use that location. We know the model on the other side is, has changed. Yeah, that one there is probably June's model. So yeah, ignore that. Just look at this area here.

说话人 3 00:06:56

按这个。

说话人 1 00:06:58

OK。

说话人 3 00:06:59

按这个就提供。

说话人 1 00:07:01

范工，您看他有橙色有黄色那几个管了吗？

说话人 3 00:07:07

嗯，看见了。

说话人 1 00:07:08

对，这就是他们现在的 NWD 里头，他们的思路，他们也是这种相当于是上下两层，只不过但是他那个地是派开的那种状态，这种的。他说如果我们有能开这个 NWD 的话，咱们可以拿这个 NWD 的位置，来做咱们的墙上的这些的规划。

说话人 3 00:07:36

我刚看了一下，存不了 NWD，咱这三维的。他能 STP 他能存吗？就是他。

说话人 1 00:07:46

他要给你发一下，然后。

说话人 3 00:07:50

他要给我发，我看我能打开，是吧？

说话人 1 00:07:53

对你看能不能打开，咱们不一定需要存那个 NWD，但只要能打开编辑就行。他给你发了一下，然后他们的 CAD 的人员说也在，拿出来他具体的位置。但这个比较麻烦，如果咱们能打开 NWD 的话，会更简单一点。然后它等于，你看它就可以用这 NWD 就可以把 STEP 文件给插入进去，咱们咱们那个就是 STEP 的文件，然后插入进去，它就能，就等于是能给它有一个这些白色模型，应该是咱们的 STEP，我理解啊。

说话人 3 00:08:37

是。那他发我，我转一下，我看能不能和他这个合上，最好就是就是合一下。而且我还看他这个，我还想问一下，刚才看他这个模型，他那个5个 TCU，那个设备是可以。随意换位置的吗？还是说。它是有摆放顺序的，还说还可以根据咱们这边。这不一共10个管口嘛，10个管口那是5个 TCU。那个位置可以调吗？

说话人 1 00:09:00

就5个 TCU。问问他。换位置是吧？

说话人 3 00:09:04

对，他能够根据咱们这边调，因为因为咱这边空间有限嘛，我那个口只能比方说第一个 TCU 只能出的第一个，第二个只能出第二个，就是他这个位置，我那个位置，要是他那边调一下，就是他这管就能对咱们这边来。OK。您就先问问他那个 TCU 的位置能不能调，还是他那 TCU 已经固定了那那5。

说话人 1 00:09:24

Okay, and yeah, we understand. And the shuangshuang will. I'll try to, if he, she can have some method to open the NWD file. And another question is about the tcus, oh, location. Cuz we found maybe the first two you're showing is the first one re corresponding to the first one, PC TCU.

说话人 2 00:09:51

They maybe not that suitable for our, the reactor collection. So it can be changed on location or it's fixed. Now we, we are you talking about how these pipes go into the TC use? Now we've already moved these around to try and line up with the piping. So it's not a problem when we finalize what's on the skid and the sequence might come through the wall, then we can move the TCUs into the right location. Okay, so it can be more around. Okay, got it. Yeah, we, the those two TCUs are the same and those three are the same. So we can, you know, it's only a matter of making sure the big ones are in the right location and the small ones, yeah, and we can do that. We need to small on this side. We can do. That's fine. Okay.

说话人 1 00:10:50

范工，TCU 是可以随意移动的，他现在只是画个示意，以后我们确定了，根据肯定是根据个咱们的那个撬块。对，撬块怎么布的方便，他 TCU 在换位置就行。他等于三个小的是一样的，两个大的你确定好位置以后，就 OK 了。

说话人 3 00:11:12

嗯嗯，好，那我明白了。耶。

说话人 2 00:11:16

Peter, the important bit for us, the penetration through the wall here is critical. There's not much room through that wall.

说话人 2 00:11:26

一个房是多少规模？1000，对吧？

说话人 2 00:11:29

And we obviously needed. There's something in the way here. And there's something in the way here, which is why we set up and then come through. So I would have said pretty much all of this space here is available, but we need to come through the wall at this height.

说话人 1 00:11:51

Okay, got it. 它的限制条件主要是，你看它 TCU 过第一道墙那块，它现在鼠标那块，跟那块有挡着的东西，所以它必须得从那块穿墙。穿完以后，它，你看它不是有一个拐弯吗？那拐弯那块也是因为有挡着的，它必须得往上拐一下。后面的这些部分，相对来说就，上边那个空间相对来说它们是充裕的。就是他穿墙的位置是非常必须的，他拐一下那块也是必须的，就这种情况。

说话人 3 00:12:35

嗯，明白了。

说话人 2 00:12:41

Yeah, we understand the other bit. So we have flanges here. We would have the wall plate to cover the opening through the wall. And then I'm guessing we would have flanges on this side. For us to connect our pipe up to, so the wall plate would go flange connection to a flange connection, we would struggle to get in here and weld anything, but we can get spanners in to do a flange up.

说话人 2 00:13:17

I noticed on your model, you were showing the vent valves. I think in this sort of location, we only need one, 1 place to vent. And I don't know whether it makes more sense to vent on this side or whether we put the vents on this side. My gut feel is we were thinking about bringing some small ball pipe work across and having a set of vent valves maybe sitting in that location. And we also need to get relief valves in here somewhere that I didn't know whether we would bring the pipes across and maybe have some reliefs and vents. In this location. So there's, it's not clear cut yet where the vents would be located. I don't think they'll be on your side of the wall. I think for access, they tend to be on this side of the wall. Got it. Because if you put 10 vent valves here, if we need to vent air from the system at this point, I don't know how we do it. We'd be climbing across the top of the skid, which doesn't really work. Okay, got it.

说话人 1 00:14:47

还有就是排气阀，我看他说的意思，咱们这块有10个排气阀放在了咱们的撬块里，他觉得，他现在鼠标指的位置，差不多有排气阀，就不是泄放，是排气的阀。

说话人 3 00:15:06

然后你勾点排气啊。

说话人 1 00:15:08

他觉得那个排气阀不应该放那块，他，咱们这个是需要一个排气点就够了，他觉得是放在现在黑色的墙的左边。那些的排气阀。

说话人 3 00:15:22

就是。

说话人 1 00:15:25

他如果这个，他如果你放在这块的话，在翘块里的话，他你再往外连的话，他不知道怎么连了，就是要排的排到外头去的时候，他可能就从这个块往上走，怎么样的？他觉得你这个排气阀得放到那个黑色的，墙的左边，就相当于是这个绿色的那群管的下面一点，那个位置。他觉得是更合适的，他能，他排放气体的位置是比较对的。还有泄放，好像也是类似的想法吧，泄放也放在这个黑色墙左边。

说话人 3 00:16:10

你不是这个黑色的墙就是左右，是吧？这不是左边这好多管，然后右边是咱的设备，是这个黑色的墙吗？对对对。中间这个竖着的这个。

说话人 1 00:16:19

对，中间那个竖着这个，嗯。

说话人 3 00:16:22

这个排气阀当时想的应该是放在管路高点吧，嗯，然后我觉得那天看了看，好像就是咱这个，它连的是管路高点，它也可以放到它那边。就是放到。墙的左边嘛，他确实是可以放那边。

说话人 1 00:16:34

他他觉得是得放在那边，放在咱们那里头反而不好弄，对于他来。

说话人 3 00:16:39

对，他应该要说连出去可能不好连。

说话人 1 00:16:41

对，他连出去很难连。我我确认一下，泄放的阀，泄放阀现在放，咱们放在哪了？

说话人 3 00:16:48

泄放阀我没听明白他的意思，泄放阀咱们没有泄放阀，咱不就有两个，就是。

说话人 1 00:16:54

安全泄放的那个。对对对。对，那个。

说话人 3 00:16:57

安全泄放排气管，我我其实还有个疑问，他安全放气排气管，他那个 PID 图里边啊，现在是相当于。是两个口，就是，要不我再展示，我看一下模型还是怎么着？

说话人 1 00:17:13

行，我我我主要是要跟他确认一下安全泄放那块，我没听清楚，他到底是想让咱们。我那我先跟问他一下吧。啊，我就问。Line of this.About the one wall and one line.Shang Shang, understand.We can put it on the left hand, left side of the black wall.Yeah, in the drawing now.And then location, he, he, she think is more suitable.And.I, we, I didn't hear it clearly about your suggestion on the relief pipeline.Is there some modification to the relief you're suggesting?Do you remember we took the reliefs out, you had the HTF relief valves on the skid.

说话人 2 00:18:05

And we move them into our side of the wall.It's just where we put those relief valves.Oh, okay.So it's our problem. Okay, got it. So the skid design is still.The the same.Okay.Yeah, if we come through the wall here, it might make sense cuz there's an isolation valve that we need.As soon as we put an isolation valve in on these returns, we need the relief this side.And it's getting to the bit, if we can't put the reliefs in until we get back to the TC use, we start looking at it thinking, is it really worth putting the isolation valve in or do we just drain it?Back to the expansion tank.So we haven't, we can't isolate the jacket easily, but we get away with the relief problem.So yeah, we haven't worked out yet whether we may end up taking the reliefs out and just have it free venting back to the expansion tank.But that's for us, Peter, not for you. Okay. It's alright. Got it. Yeah.

说话人 1 00:19:28

yeah, got it. OK, 范工，不是咱们的问题了，是他在想他们那边，毕竟他那个泄放那块是到他们那边了，他们那边怎么排这个泄放的问题了？咱们不用特别管，咱们那块泄放的事，咱们泄放管已经不在咱们那块了，泄放那个阀，那个隔离阀已经不在咱们这块了，所以就让他来想就好了。

说话人 3 00:19:54

嗯，好的。

说话人 1 00:19:57

OK，那我就继续往下了。

说话人 1 00:20:02

Okay, so the, sorry, so the second question is about the SEO one and Cuo 1. We're proposing to make it against the wall and so that we can have the space for the.

说话人 2 00:20:25

Well, maintenance and withdrawing the level gauge, I don't think we will be able to get into the skid. So if we had to do maintenance on those. I think we would drop the reactor subassembly out, pull it out so we can access those bits of equipment if we can. And if we can access from the right hand side, there'll be some access we can get from there. But I don't think there'll be too much. But yeah, we have to do significant maintenance on those then. I would say we would have to pull the whole reactor subassembly out. Yeah, which when we do the installation, the reactor subassembly will come without the control panel on there. We bring it to site, we put the control panel on and wire it back into the control panel if we needed then to do maintenance on the sub assembly behind it, we could just pull the whole sub assembly out. We wouldn't disconnect the instrumentation. So it still makes good sense to put the instrumentation into panels on each sub assembly. To make maintenance easier. But yeah, I kind of envisage we would just pull that whole sub assembly out, disconnect the pipe work and maybe a few of the instruments if they go between the two to give us access in there. Okay, because we'd always be reaching and stretching and some of those flanges, the class 900 flanges are heavy. And if you've got awkward shapes to lift as well, you. You might be a two man lift or you might need some lifting aid. Yeah, I when we get the layout, we can look, but I struggle to see how somebody would squeeze in there. and actually do maintenance.

说话人 1 00:22:48

OK, understand。那个范工，他还是以就是把分鞘给拉出来，再做维护和抽取的工工作为主。比如说他想法在让这些仪表还，也就是仪表也在分鞘上，他有一个你的控制柜，相当于是在他分的控制柜嘛，控制柜也跟着这个分鞘走，不需要拆仪表，只是拆管就行了。他把那个分鞘拉出来，他如果你这边有你说的，他觉得这个抽取空间应该还是非常有限的，因为他好多这个900的法兰还是很重的，他觉得挺难的。如果你这边有现在比如现成的已经这个思路可以给他看看，能不能行呢？你现在是有吗？还是说就已经布完了？还是说还没有呢？

说话人 3 00:23:55

就是我是按照上次他建议的我还没布，但是我现在是按他布那个布了，布了之后我发现。就是一些阀和那液位计啊，现在都靠墙了。你把前面那个撬抽出来之后，然后后边那几个阀也够不着，所以我想让这个 SE 01和 CU 01和。和这个阀的位置换一下。我要不你再共享一下。

说话人 1 00:24:21

行，你共享一下，然后说一下，啊，我明白了，就是你即使抽完以后，还不太行，是吧？

说话人 3 00:24:27

对，因为他抽完了之后，这个S101和 CU 01他，他上次不是建议往前放嘛，他就正好把那个挡上了，你等我那个那什么看一眼就知道。

说话人 1 00:24:39

OK。

说话人 4 00:24:41

哎，李博，我刚听到他说这个，他维护的时候把鞘分开之后，那个仪表线可以，现在可以不断吗？我我没太听清楚啊。对，我听的是这个意思。一会说不能断的，——一会说要能断，能能不断，一会又说好像还是要弄什么的。

说话人 1 00:24:58

我听这意思，他说每个鞘上都有自己的 pane，这样的话，他的仪表就可以不拆下来，直接就断点管就行了。

说话人 4 00:25:10

有次的 pane，他这样，他每个 pane 都是需要有，哎呀，他。

说话人 3 00:25:21

那我我觉得。

说话人 4 00:25:22

影响情况了。

说话人 3 00:25:24

那我记得他说的，我我那天想了一下，如果咱们不是要抽这个鞘吗？你看我屏幕哈。抽这个鞘，咱们把这个鞘上所有的仪表都放在一个箱子上就行了。他是这个意思吗？比方说你剩下的这两个鞘，你是不抽的。这两个鞘再给他放一个，我是这么理解，他再放一个这个，就那个。对对对，我觉得他是这个意思，是到原来咱不说一共就一个这个吗？现在是不是要分成两个？一个和他放在一块，他跟他抽，另一个是这两个鞘的一个，就是他就可以不动了吗？他，case 是这意思吗？

说话人 4 00:26:04

对，这，哦，这个这个当时我们还是，现在还是没这样去分，去去弄的。那一个是我们看看你有没有办法去分。再一个是他的想法是挺理想的，因为但是他实际的这个恰好这个东西接连接起来之后，他可能不止一个箱子，就可能是两个或者什么的箱子，因为这恰好如果他连接了，比方说他肯定还会有会有几个这种什么硬连锁啊什么东西，这种东西。他那东西都是都是连在一块的，他做不到，当前的这个结构，这些信号类型啊，这类型啊，他做不到，他只有一个鞘就把他全包了。

说话人 3 00:26:50

您是说这个，就是比方说咱们抽的这个鞘里边，它并不全是那个什么 IO 点，是吧？还有别的点，别的类型的。

说话人 4 00:26:57

它并不，并不全是进这个箱子的点，还有可能会进别的箱子，就是有有可能还会有小的，比如说要小的一个一个转接，非本安的一个转接箱，转接盒，或者本安的一个转接盒，这种它会有。多个东西，再加上你可能。因为我们现在总共是有几类的东西哈，一个是。非本安的转接盒，本安的转接盒，加上这个本安接线箱，还有一个触摸屏，或者灰膜屏，或者是那个按钮，也会去到到他中控室去。

说话人 1 00:27:37

那就是原原则是，就是说这些线尽量少，然后少到一定程度就行，那你做不到完全没有，但也可以就是说尽量少拆这些仪表，的连线，也也行，我觉得。

说话人 4 00:27:54

那少拆，那倒没算过，但是不知道会有多少线。倒是多，倒是也不会太多，倒是。

说话人 1 00:28:00

嗯，到时候再看吧，到时候这个方面也再跟他确认确认。那行，那意思就说，我们还是要分成两个鞘了，基本上是要。一会再再细给他。

说话人 4 00:28:13

两个箱子了，可能要。

说话人 3 00:28:14

现在是现在是三个鞘嘛，然后。三个鞘。就是这个鞘是抽拉的嘛，他到时候要把它抽出来。但是。

说话人 4 00:28:22

也就这个鞘单独单独做箱子，对吧？

说话人 3 00:28:25

对的，这个鞘单独做箱子，跟他一块走。但是您说不。不光这个 IO 点嘛，你不说还有别的点嘛，是吧？别的点。

说话人 4 00:28:33

对，就不光不光只接这一项的，还有别进别的地方的一些有有线来。八成会有，我现在没有说，没有把它所有的线都分开，信号都捋清楚，所以不好说有没有，有多少，或者有没有。

说话人 3 00:28:48

那要不我下来就是给你分一下这里边有哪个点，就是现在以我现在这个布置上面，就是哪个点是归到这个壳上的，然后你再看一眼是有多少个线和它和外，除了和它连，还和别的连。

说话人 4 00:29:00

是的是的是的是的，所有的不是控制设备都都要给我们一下。

说话人 3 00:29:04

但是孟工，我上次又说了，您不说他这个也不会特别小嘛，分成两个，比方说是吧？

说话人 4 00:29:09

不是这样，他那个箱子，他不是那个 case，上次不是说我找不到小的，他帮我改的。

说话人 3 00:29:16

他说的那是一，他上次我理解他说的还是一个箱子，就做成一个箱子。

说话人 1 00:29:21

不是不是，那个是另外的箱子。

说话人 4 00:29:23

对。对，这个再说了，他他如果他如果能如果能找到可以小点的箱子，那我们分开分开再说了啊。对。行。就是他的目前来看，我们因为什么，厚度450，这个变不了了，基本上因为他的结构就在这。厚度和高度，单，高度是。这个。

说话人 3 00:29:46

不是330吗？高，高度。这个。

说话人 1 00:29:48

高度400。高度400。

说话人 3 00:29:49

对，高度400。

说话人 4 00:29:49

500，300。

说话人 1 00:29:50

我打断一下，那个咱们得赶快，就是，因为他在等，然后咱们得赶快，先把这个事给一点点说，好吧？那就我们尽量自己少讨论一些，啊。然后就是你您您说，啊。

说话人 3 00:30:07

我我接着说，就是，这是按照上次 case 说的，把这个。分离器和冷却器放到前头，它后边不是还有好多阀吗？我是认为它这如果这个，这个壳往外抽的时候。那这些个阀在后头，它是不便于操作维修的，如果把这两个，和这些个阀换个位置。我觉得就好一些。

说话人 1 00:30:32

行，OK，我就这么说一下。Hi, Keith.Yeah, yeah, we found the left hand side is the, we saw the pullable skid and we can drag it out.And now we're following your suggestion on making the two equipments nearer to the.To the skate, but the valve and the transmitters needed to be maintained is on, behind the equipments and near the wall.And so if we're following this one, even though we drag it, drag this sub, skate out, and then it's also hard to reach.The equi, the instrument that need to be maintained.So we're thinking to switch the precision of the valves and transmitters and the, the, the equipments, the two equipments, the green one and the

brown or orange one. And so that when we're pulling out the sub skate that we can easily access to the. To the maintenance point and do the, do the operations, I.

说话人 2 00:32:00

So you're thinking of swapping those two, in effect, putting the condenser, the green one. Yeah, the condenser. And into the middle more, and the separator on that end?

说话人 1 00:32:17

And making the valve and the transmitter to the front so that when we pulling out this sub skill is in the front and we can operate.

说话人 2 00:32:39

Yes. Can we not reduce the width of the skid? Have we got free space in the middle of the skid that we could reduce rather than have that assembly at the back being so wide, can we reduce the width? I think the bit that will dictate how wide that is, will be looking at the BTO1 and how wide it needs to be there because we've got BTO1 going down to the pump in front of it. So which bit dictates the actual width of that skid? Is it the BTA1 in the pump in front of it, or is it beyond?

说话人 2 00:33:22

I would put the separator, get the reactor bay in front. The reactor bay looks pretty tight. You know, there seems to be a bit of space. Okay. If we move, say, the reactor skid back by 1 or 200 mill. Does that then get rid of the front back question on the separators, cuz they'll be about central, and does that then allow us enough room to get BTO1 and the pump in front of it on the other end?

说话人 1 00:34:02

OK, let me ask.

说话人 1 00:34:06

那个范工，我理解他的意思是说我们现在的宽度的限制，是这个罐的 BT 01 的罐的宽度限制，还是还是哪的限制？他在想是不是我们把整体的后翘的那部分的宽度减小了 100 毫米，或者 200 毫米的，是不是就能解决前后问题了？前后不好操作这个问题了？因为当时不是说当时留空间嘛，在如果不留空间，因为是要把给拽出来的嘛，不留这个空间，是不是它能更紧凑一点？能够能够解决这个问题。他，我听这意思好像是从这块穿过去，直接拿手穿过去，少个一两百毫米以后，我就可以直接拿手穿过去了。意思吗？

说话人 3 00:35:16

不是，这边反而还挺多的。而且还有一个问题，就是你看这个，如果我把它放到这个上面的话，放到这边的话，它顶上就是 TCU 的管，它到时候抽是抽不出来的。但如果我要放在这边的话，这顶上这块，因为咱们是这么上来拐上去的嘛，这还有点空间。我觉得这个液位计可能就能，到时候就从这鞘里边能抽出来了。

说话人 1 00:35:40

OK。

说话人 3 00:35:42

你再说他刚才说的的问题，他那是意思是说咱们这个宽度，就是他想再减一点，现在受限于哪个东，哪个？是，是不是这个 BT 01，是吧？

说话人 1 00:35:51

是，是吗？

说话人 3 00:35:53

他是，他不就，他是这个意思吧？他是说。

说话人 1 00:35:55

受限于哪个，他不太确定。

说话人 3 00:35:58

他就是想，还是想把这个鞘再整体做小点，对，就宽度做小一点。

说话人 1 00:36:02

对，宽度做小点，原因，这样的话就。你相当于是把前鞘给抽出去，你就不在乎到底是在哪边了，他是阀是在前还是在后嘛。

说话人 3 00:36:16

你抽出去之后，他要是在前的话，这两个在这挡着呢，他这后边他还是没法下手啊。

说话人 1 00:36:25

对，我我我也没听懂他为什么减宽，为什么就可以解决这个事了？

说话人 3 00:36:34

他，我还，他没说把这个阀挪别地去吗？那现在也没有别的地，只有这个地了啊。先说他这个，他说的这个吧，这个。这个的宽度，这宽度好像是800，我记得是一共是1米5嘛，前面这个撬是700，后边是800。再减100，那我觉得100的意义也不是很大，100利益700人伸手，你要是想拧那个阀，也是拧不了的。而且这个，这我觉得已经挺紧张了，他再减，主要是这边，您看。就这个，因为这个泵吧，也是我上次他们给的那个模型，就像第一次问的时候有一个模型。再减的话，再往前错，这已经没有量可以。就这宽度再减的话，这个泵都不好装了，还有这，他因为他这出口有个阀嘛。

说话人 1 00:37:29

所以就是，我觉得，泵加这个 BT 01这个是它的宽度限制。

说话人 3 00:37:35

宽度限制，对。再再往后减，减不了一百。

说话人 1 00:37:38

OK。

说话人 3 00:37:39

嗯，而且您再给他说一下，就是这个液位计，到时候我认为放到这个，靠墙的这块，顶上这可能还有空间能抽出来。如果要放到这边的话，这个 TCU 这个管肯定是挡他往外抽这个液位计。

说话人 1 00:37:54

嗯，OK，行。

说话人 1 00:37:57

hi, Keith, we, I just. Ask about the bottleneck, and it is the pump and BTO1 and the valve. And we thought it may be hard to reduce 100 millimeter because of the pump and the BTO1 and the valve has. Already been a little bit narrow, and another reason that we want to swipe the position of it is about the level trans. The transmitter we saw on the top of the orange one and the it is be blocked by the TCO pipeline. On the head. So it maybe not, it maybe not able to pull it up, but if it's going to against the valve, then it could be, there are more space there and it could be dragged out up. And also Shuang, Shuang is a little bit confusing on if even if we. Shrink like the shrink the whites by 100 millimeter, we still cannot access to most of the walls from the front to do the maintenance for the cleaning to, for the maintenance and.

说话人 2 00:39:36

To reach the valve, there aren't that many valve.

说话人 1 00:39:44

Around the back, the only valves. Think of the CIP.

说话人 1 00:39:58

Yeah, but at some time we need to reach any of the wall in some situation for maintenance purpose or something, don't know.

说话人 2 00:40:14

I'm not sure there are that many valves around the back or they don't need, it doesn't need to have that many valves around the back. Okay, that's a thought, not a no.

说话人 2 00:41:06

So we have a couple of valves around the polishing filter, which we could put. I think we could probably get some access from the end if those valves were accessible from underneath the sampler, as the pipework then goes out towards the condenser.

说话人 2 00:41:35

And then we got the valves that come through to the sampler. So again, those, I think will be accessible from the end.

说话人 2 00:42:00

Oh, I think this is an old P and RD I'm looking at, to be honest. There's some valves going from the separator to BTO4. Those may be. We'd need to look at those. There would be small bore as well. I think we. I think we have a look when we've got, are we in a position where we've, everything is pretty much there if we swap those around or we moved it, so we went to the condenser that's more towards the middle of the skid and the separator on the end. Are we pretty much there with everything in a proposed position? Cuz the thoughts are when we get it pretty close, we then start going through it and we look at what's, what doesn't look good. That's a relative to what doesn't, what looks like it could be improved and see whether there's ways of improving it, whether the valve actually offers any benefit in that position.

说话人 2 00:43:37

Yeah, you know, it might be a case that I think on our pipework on the reactor skid, where we have the valves that either use one reactor, two reactors or three reactors, those manual valves. I think in a lot of cases, we've got valves that are pretty much next door to each other. And we could delete some. So I'll, I think there's some rationalization on the valves we could do. Yeah, so, okay, can't access a valve. We could always pneumatic actuator on it and just run an air supply to it so they can open it manually from. Say, the end of the skid. If you look at that back corner of the skid, so on the, oh, you've got the sampler on the end of the skid. If you go to the back of the skid there, you could have maybe half a dozen small valves on an air supply. And you could run some actuated valves, just put actuators on some of the valves for the CIP if you can't access them. So you can open them remotely, but it's a simple valve arrangement that. That may work well, especially on the CIP where we're talking half inch comment where it's half inch or three quarter inch. I think actually you might have said 10 mill last time. It might be worth keeping them at half inch so we can get some mainstream valves that are easy to actuate. Rather than going for the sway. Yeah, got it. And then we could just couple of little 6 mill valves on the side there that run an air supply to the actuators. So yeah, if we find those valves that aren't accessible, there are ways that we can.

说话人 1 00:45:43

Actuate them so they're accessible to the operator. Okay, got it. And how about the the level transmit the transmitter I couldn't remember is temperature maybe temperature we need to move it a little bit back and to giving space to put that.

说话人 2 00:46:05

Yeah, and like you say, maybe it does make sense to swap the position. So that's on the end of the skid. Yeah, so we can access it from that end. And there's not much the condensers, just a vessel. So we very rarely need to access the condenser. Yeah, so yeah, but that probably makes sense. The relief for the condenser, we could. We could probably access that from some steps. So that was put towards the top of the skid, we could take that off for testing.

说话人 2 00:47:08

And then the reliefs for the reactors, I've not looked to see where they are, but I'm guessing they are accessible from the front of the skid. Is that it? We've got four high pressure reliefs. Yeah, so maybe.

说话人 2 00:47:36

They just need to be accessible from the front for the reactors, and maybe the end for the condenser.

说话人 2 00:48:05

Feed in there.

说话人 2 00:49:37

The relief valves, the high pressure ones, they're all the same size, aren't they? Need a separator, condenser. They're all the same size reliefs.

说话人 1 00:49:54

范工，高压的泄放阀，他们是都是同大小的吗？这四个，分离器、稳凝器这些的，泄放阀都是一样大的吗？

说话人 3 00:50:14

泄放阀是是安全阀吗？

说话人 1 00:50:17

对，安全阀，对，他们的泄放是安全的。

说话人 3 00:50:20

对对对，都是一样的。

说话人 2 00:50:22

Yeah, it's all the same size. Does it make sense to. Let me share my screen. Okay, sure.

说话人 2 00:50:56

And that's not the latest PNID, but it hasn't changed that much. We have this relief off of the separator, off the condenser, rather. If they are that close, could we pipe that relief? Into this line. So we use that one relief valve rather than two relief valves. So we're just up a line into there because in theory, I'm guessing, is this is on, is this on the outlet from the condenser? So that relief was on the outlet. It doesn't matter where it's there or whether it releases from there, you could either little relief bypass over that valve so if it was closed, it would still relieve through and out, or we could just pipe it straight into that relief line and use that relief for both.

说话人 1 00:52:06

Okay, so combining PSV 0,2,5 and PSV 0,1,9,1. And then into location, we put the relief out somewhere where it's accessible because one of those would be on the end of the skid.

说话人 2 00:52:32

Cuz in theory, that valve is just gives us a bit of back pressure. So those, those, there is no valve between those two. So we didn't need that.

说话人 1 00:52:46

And we would be fine. Okay. So you mean you say we didn't need that is you mean we don't need the back to back pressure while we. 9,9,6,9.

说话人 2 00:53:05

You we need to delete it. If we didn't need it, then that would just be a bit of open pipe and we could put one relief to protect the system. Okay. So that, that's causes us a little bit of a problem to only put the one relief it, but we could go from here to there. Oh, it's probably not as tidy anyway. You could put an orifice plate in there and it just gave you a little bit of back pressure. So you can't actually close it. That would be another option. But to be honest, that gives us more flexibility. That's. And it's easier to install a little 10 mill needle valve. I would have said just pipe that into that line and then you've only got one relief. You can put that relief valve towards the edge of the skid so it's accessible for maintenance.

说话人 1 00:54:07

也。

说话人 2 00:54:08

what does shankuan shankuan think?

说话人 1 00:54:12

诶, let me ask, 那个范工。啊, 他那个主要想法就是几个, 他主要想法第一个关于这些阀门, 他说可以很多, 你觉得够不到的地, 他就改气动, 改气动完以后, 相当于变成气动的那种, 就放一个气动气的这种的, 这样的话, 那些阀就相当于操作工还是能。开关, 就比较方便, 这些的, 另外他还有一个想法, 现在这个 PSV 025。

说话人 1 00:54:52

跟 PSV 019，实际上在3D图上，它们离挨的是比较近的。是一个是这个分离器的，一个是这个冷凝器的泄放阀。它们俩其实可以合并一下，就连一下，连一下以后都变成一个阀来泄。他觉得是这样，连一下应该可以，就变成一个阀门来泄放。这样的话，阀就025跟019合并的这一，就是等于是消除了一个阀，另外那个阀就可以变到一个。比如说靠着墙的位置，好够到的位置，这样就能解决很多阀门的问题。

说话人 3 00:55:46

李博，他还是不建议我换过来位置，是吧？

说话人 1 00:55:50

对，他没有特别建议那个换过来的位置，我跟他说就是。探针那块，探针那块他说，他确实觉得是得把给放后头，把橙色设备给放后头，这样探针能拉出来，放到后头一点，跟他错开。

说话人 2 00:56:14

Oh, Peter, oh, hold that thought. That doesn't make good sense cuz all the flow will go that way, wouldn't it?

说话人 1 00:56:27

他又觉得好像不太对。

说话人 3 00:56:31

就，我觉得这俩串起来了，我觉得确实有点，那不等于这俩连通了吗？

说话人 1 00:56:36

对，他他就是这个。

说话人 3 00:56:40

他是想把这两个安全阀弄成一个？

说话人 1 00:56:44

对，串起来就连在一起，然后就变成一个，但他他又说不太对，然后那个。

说话人 3 00:56:51

不太对，那个025它是保护那个冷凝器的那个是壳侧的，不是公益侧的，他俩不能串。

说话人 2 00:57:06

Okay, it's in different side. One is what you say. That was a stupid idea. It doesn't work. But there's no reason why we couldn't put both of those valves towards the edge of the skid cuz it's only gonna be half a meter of pipe.

说话人 1 00:57:26

It's not gonna be a long length of pipe, really, is it? So, sorry, could you say that again? There was some internet issue and it's just missed. We can't combine those two. Yeah, be no, cuz if we pipe that in, all the flow would go that way. Yep, it was stupid idea.

说话人 2 00:57:55

But we could locate that valve and that valve towards the end of the skid to make them accessible. The length of the inlet pipe wouldn't be that long. Yeah. Oh.

说话人 1 00:58:04

OK, got it, thanks.

说话人 1 00:58:07

对，他意思就是 PSV 025，就跟 PSV 019，他放在墙边就行，他进出口的管没那么远。

说话人 1 00:58:21

I mean, number chart. Okay, so to wrap it up, so for the walls, we can, for the like the relief wall, we can make it against the wall. And for the other walls, small walls, and we can use the panoramic actual role. And for the, how about for the level transmitter, the temperature transmitter, we want to put the orange equipment a little bit against the wall or a little bit behind the temperature transmitter on.

说话人 2 00:59:06

Yeah, we, we can bring that to the end of the skid as well if we put the separator like you said on the end, so we swap the condenser and the separator, then we could have the Lev, the temperature transmitter on that end as well potentially. Are you talking about TT eighteen, the one on the pipe?

说话人 1 00:59:35

范工，你要不然再共享一下，把这个温度计的那块再指一下。

说话人 3 00:59:47

是液位计。

说话人 1 00:59:49

液位计。

说话人 1 00:59:50

Yeah, we did this one. The, the, I see, Linda. Yeah, we did the is level transmitter. Sorry for the s SEO one is original B blue and now we click it so it's green. Yeah, this one. So you want to this SEO one to swipe with the. The condenser, you mean, right? Or. Yeah. Which one?

说话人 2 01:00:31

It's kind of a bit of a zigzag, isn't it? So if you go from the plan, if you look down from above.

说话人 1 01:00:40

我们从上面看，往下看了一下。

说话人 2 01:00:46

So we will come out the reactor on the far, on the bottom. So we come out that reactor yet the reactor on the right, and then we would go back towards the condenser and then. So in effect, we put the separator on the end of the skid where the condenser is, and then we try and position the condenser to minimize the pipe work going from the reactor to the condenser to the separator.

说话人 1 01:01:27

So you mean the separator going behind, against the wall, behind the condenser? We're still the same location. Here's a question for you, Peter. Can you put.

说话人 2 01:01:45

The condenser, in effect, under the control panel and the separator in the back corner of the skid. K. Does that give you enough for it?

说话人 1 01:02:01

范工，咱们现在这种情况能把冷凝器给，放在现在的箱的底下。

说话人 3 01:02:13

这个。放在这个箱子底下。

说话人 1 01:02:16

对，就等于是跟这个 reactor 更近一点，把这个分离器放在冷凝器的后头，跟墙之间，等于是它是这样的
的一个 L 型，够够空间吗？

说话人 3 01:02:38

我还没太听清楚，冷凝器放在。

说话人 1 01:02:41

这控制柜的底下。箱子。对，对，就等于往前走点，刚才。

说话人 3 01:02:47

往前走点没有空间了，这马上就是这个阀了。

说话人 1 01:02:53

他就只是说没说一定要这么放，说他能不能这样的话，有位置把那个分离器给放在给这个墙跟墙跟中
间，墙跟叫什么？冷凝器中间，就等于它现在不是平行的嘛，把它变成。把它转90度。对。这样的话
变成这样一个排的，它。

说话人 3 01:03:25

90度，然后往，是往里推吗？就是推到这里头来，是，是推到挨着挨着这个BT04吗？就是挨到这边
来。

说话人 1 01:03:33

对呀，这不是给液位计给找个地吗？找个道吗？

说话人 3 01:03:44

往这边挪，来这边，现在它，冷凝器放在分离器的位置，分离器再往。

说话人 1 01:03:51

不是不是，冷凝器往前放，我跟前头那个桥更近一些。

说话人 3 01:04:00

他放不到前那翘里边去，现在没有空间放到前头这个翘里头。

说话人 1 01:04:06

那这个分离器往后放能放多厚啊？

说话人 3 01:04:12

分，分离器往后放是说是往这个后放，还是挨着墙放呢？是挨着墙放吗？

说话人 1 01:04:19

对，挨着墙放，就不是，在咱们的目的还是错开，咱们要不是说那个，说怎么放怎么放，咱们目的不就是还要错开这个那个 level transmitter 嘛，那个那个液位计嘛，你想，你想。

说话人 3 01:04:34

还是嗯。对。那还是他要是，哎呀他的话，那就只能还是往这个挨着墙边放才能。瞅这个空过，往上瞅。

说话人 1 01:04:42

对呀，他的意思就是说，你现在，那不就是，就是你没法直接放在有没有办法直接放在，唉，冷凝器的正后方，这种状态，冷凝器往前放一点，他放正后方，没有办法，是吧？那错开一点。那没有办法。

说话人 3 01:05:05

对。

说话人 1 01:05:06

那。错开一点呢？

说话人 3 01:05:07

不是错开一点，是只有把这个分离器放在挨着墙的地，他才能够往外抽呢，这个往这边放的话。他就，这个如果往这边放的话，就是半个，他一半在这个翘上，一半在这个翘上了。

说话人 1 01:05:22

对，就是这种状态。

说话人 3 01:05:29

这样的话，他是为什么呢？是六块，这个往这边放，我明白是为了把这个抽出来，这个为什么往这边？是为了有空间。

说话人 1 01:05:38

对，有空间让他放在他后头。

说话人 3 01:05:47

他不是还是这个位置吗？只是往这边，往这个，把这个那个。

说话人 1 01:05:52

对，前翘，对，移一点。

说话人 3 01:05:54

往前翘移一下啊。对对。然后这个他意思不就是说往，是为了把这个抽出来，找一个位置放吗？

说话人 1 01:06:03

他对啊，他就是往，他往他往前移了以后，这个空间够不够？他直接往他的身后，就放在冷凝器的身后，有没有空间？

说话人 3 01:06:22

他说的身后是指放在这个位置吗？

说话人 1 01:06:24

对，放在这个位置，是的。

说话人 3 01:06:26

哦哦，那这，他为什么要这么。放，这个面不就，那这个面不就堵死了吗？相当于你这个往这边走了，这个往这边走了。就相当于现在是，就是现在他俩是这么排，然后转90度之后，这个往这个。嗯，我明白他的意思了。

说话人 1 01:06:47

你看看这个能，一，一是能不能放，二是你觉得他堵死了，他这个卡点在哪块？然后就为什么觉得这个不太行？

说话人 3 01:06:55

也不是，也不是说堵死了，因为他他要把这个撬拉出来，堵死了也没问题，我我是总想着。如果这两个都同时挨着墙的话，中间有个道可以进去，有的时候可以不抽这个鞘。但是他这么。

说话人 1 01:07:09

他很坚决的说一定要抽鞘了，咱们就按。一定要抽鞘。这种的想法走呗。

说话人 3 01:07:16

等一下，我看看他有，我看。不抽鞘。我看看他有多大。

说话人 1 01:07:20

ok。

说话人 3 01:07:22

我我我来这个电脑上看一看。他这个直径是400多，放应该是能放下，就是如果这个就是一半在这边，一半在这边，把这个。把这个挪这来，可能是能放下的，但是放下来之后呢，中间不就是空了吗？是这意思吧？这个往这边挪，这个挪这来。

说话人 2 01:07:56

Peter, looking at the end of that skid, it looks like you've got the HTF pipework coming out from reactor 3 and then it doubles back on itself. This is where getting the model layout looked out helps with the reactor design. Surely that nozzle should come out the other side, maybe a 45 from the other side. So it pushes back towards where our pipes go up. And then that frees up some space around there. To put the small ball piping in for the condenser, we're not being constrained by nozzle positions on reactors that can still be moved our way, no. Because that takes up it. It just makes the pipe run untidy. See what I mean? If you look down the bottom of the reactor 3, you got the pipe work coming out towards the end of the skid, then it goes across the end of the skid, then it goes back into the middle of the skid. Before it goes up for the transition through the wall, if that came out the other side of the skid, then you free up all of that place. There's a fair bit of wasted space there, okay? And that's kind of where we make the skid work by getting rid of all of those small. Small bits of wasted space. Oh, okay, got it.

说话人 1 01:09:57

And about the previous one, I just ask Shuang Shuang, and she thinks, yeah, we can move a little bit forward of the condenser and then put it, put the separator on between the condenser and the valve. As you said, there are enough space. And so. So that make it work. But the condenser then will be in the middle of these 2 sub skate. And yeah, and what the space for the current separators location w, w will be use for.

说话人 2 01:10:45

That, I kind of get the feeling that's gonna give us a bit of a hole in the middle of the skid, but that hole wasn't really accessible and it might be a perfect place to make running the HTF pipe work easier. You know, we, if we used it for some fully welded pipe work. We'll never need to access it, and it could make our HTF pipe runs a little bit easier to install. Okay, okay, got it. Let me trans the transfer this to Shawn.

说话人 1 01:11:26

first。那个范工，他说了两件事，第一件事就是我问了一下他这样的话，中间就有一个空间，就现在这么放完以后，分离器现在的位置，他就等于空出来了。空出来我说他用这个空间干嘛？他觉得可能就是这样的话，这个 TCU 的管路这些可能就变成可以变成这个，就是全焊接的了，就更好排这个 TCU 的管路，这是一个他觉得的好处跟方便点。另外就是现在你看 R3，反应釜三，那块的 TCU 管路，他这块因为管嘴的问题的，朝向的问题，他的意思，他现在就是曲曲扭扭的，他觉得可以通过改变这个管嘴的朝向，再优化一下，它就不必这么曲曲扭扭的，也会省下很多的空间。

说话人 3 01:12:32

这个 TCO，它现在认为这条管就是跑的有点弯头太多，是吗？

说话人 1 01:12:41

就很占空间，就是它先往。

说话人 3 01:12:44

它现在也，我觉得。因为它上边有一个温度变送器，还有一个排，就是相当于一个排净阀。没有别的位置，如果我又把它扔到正这个方向来。这会占空间太长了，只有这么拐两个弯才能把这点量拐出来。

说话人 1 01:13:06

他给了一个方案，太长了，刚才就是时间也比较久嘛，我得重新听一遍那个方案。他给了一个方案，说管嘴从哪边开，怎么样的。这，才大家有没有翻译器能听得比较懂的方案的？谁能帮个忙的？林博士，您在吗？您刚才听懂他那个方案是啥了吗？

说话人 5 01:13:38

哎，不好意思啊，我刚刚没听，我就为了卡住一个一个东西。

说话人 1 01:13:42

啊，行。你再说一遍。那没事，就是他给方案，大家都没听的话，我就回去我自己再重新听一遍再录音就行。反正他就是这个想法。不好意思啊。没事没事，其他人有翻译器里头当时有什么说法的吗？没有的话，那就我到时候再听吧。范工，您继续想跟他怎么问？比较就是您的思路跟想法是什么？

说话人 3 01:14:24

我觉得我可以按照他说的摆一下，把我想摆的也摆一下，对比一下，看一下。就是老这么说，可能。可能都说的比较那啥，就是，就把我想摆的位置给他摆一下，然后他刚才说的。顺，横着放的位置也摆一下，然后就能看出哪个更便捷一点。

说话人 1 01:14:47

行，OK。行，可以啊。耶。

说话人 1 01:14:51

I just talk to Shuang and we, she thinks, yeah, we can do that. And she, she, she, what she thought is the next step could be she will like following the idea that. Making the condenser a little bit forward and making this separator behind the condenser and against the wall, and then making the space for the TCO pipeline and for this plan. And also. Doing the optimize of the nozzle. So freezing, freeing up some space for the TCO pipeline on the R reactor 3, and she can do this plan and to give out another version of the 3D model. And also. She want to also illustrate her plan, like SW swapping part of the condenser. There and separated against the wall, and we can compare which one is more convenient and which one we often as some more steps could be easier plan. So that, yeah, that's good. Discuss more clear cuz I'm translating the plans to her and I may understand it. One way. Yeah, she maybe the still a little bit confusing. So we combine the idea and making out the 3D model and discuss on the 3D model. Maybe easier. Now, Shuang, Shuang, Shuang, when she's moving it around in the model, she'll get a much better feel for what's possible. Then, then what we will just looking at it. Yeah, we come up with ideas, but it's not until you move it in the model you see whether it really could work. Okay, thanks. So we will come back to you about these plans later after Shawn got the. Move that around and finalize some plan.

说话人 2 01:17:13

yeah, no, that's perfect.

说话人 1 01:17:15

OK, thanks. 行，我跟他说了一下，按照这两个咱们都排一下，给他看看，我们再深入讨论一下，你看行吗？范工。好的好的。我明天的上午吧，我再听一下他的具体在3的那块，反应器3的那块是怎么？想怎么改的？管嘴的方案，我要到时候给你发一下。

说话人 3 01:17:41

嗯，好的好的，李博士。

说话人 1 01:17:43

OK，好。

说话人 1 01:17:45

Yeah, so do you still have some time, like for half an hour and I'll get that work 15 minutes.Yeah, sure.Thanks.It's always the guys that need to go home. Yeah, it's the.It's the third question and the other questions maybe faster.And the third one is for the operating the CIP and the drain lines from the front of the skid will increase the delay. Shuang, Shuang, could you share the screen and.And explain more.方工，这第三个问题，您那个再共享一下，我们你再。

说话人 3 01:18:40

就是上次 case 说，把排净的都挪到前头来，咱就以这个，这个分离器为例。它如果咱把排液挪过来，那这么长的管路不都是到时候会存着液体吗？OK。这个，我看，这不挪到前头来，因为它的这个比较靠后的位置嘛。要这么挪过来，这个。它有。

说话人 1 01:19:05

Last time we talk about the drainage and you, we were saying that putting the drainage to the front and but we found shuangshuang trying to put it as the as here, showing here and put it in the front. But she was thinking that.If we doing that, the length of the pipeline increases, and there are a lot of that like.That can reserve the, the, this solvent.And so what do you think?Yeah, I'm with you on that one. What was one moment? Is that off the separator?

说话人 1 01:19:55

这个是从分离器上走的， yeah。

说话人 1 01:19:58

The separate room. This one. Do it.

说话人 3 01:20:00

就这分离器，就这条管，就这么过来了，这是那个排定阀。那这条管。

说话人 2 01:20:08

The this one is for the separate. Okay. If you hold the model there and just go up a little bit, we'll shine there. This might be, yeah, so on that to be on back right and still. Just a thought. If we had some, to be honest, if we put the separator at the back there, we can probably access the valves from that end, can't we? I agree. The valve should be close to the vessel. So let me share mine. Okay. So we have the drain valve. So I'm visited.

说话人 2 01:21:07

We come out the bottom of the vessel. We have a horizontal minimum length there and then the valve. And then bring this pipe out with the blank flange to the front of the skid for connecting our hose connection, but the actual valve in the isolation is close to the t. It's in the horizon, so it doesn't fill up with. Reaction mix or solvent, and as long as the valve holds all of this pipe to the front of the skid should be empty. If for some reason that valve lets by, then it will fill up. But we'll know that when we take the blank off, because we'll get liquid come out.

说话人 2 01:22:06

If that valve then isn't accessible, we could put an actuator on there and then just run it from a small set of small ball valves. So we just open our ball valve, which puts air onto the actuator and drives the valve open. So that was kind of my thoughts is if we can't access these, we'll little actuator on it and we'll run it remotely as a manual valve with the remote operation. Yeah, but yeah, that needs to be minimized. Otherwise, we'll end up with either solvent or reactant sitting in there that's either wasted or. Potentially gives us a problem when we're cleaning because we'll just end up struggling to get rid of the residue.

说话人 1 01:22:59

Yeah, okay. It's a lot. Let me digest it and transfer. I wanna go back the model. If you go back to the model and we look at the bottom of the separator, okay, see where the part run is.

说话人 1 01:23:23

OK, 范工, 咱们回到你的模型那块, 我们看一下他这个管是怎么走的, 回到这块页。

说话人 2 01:23:35

So in essence, the flange at the front of the skid stays where it is, but the valve moves. Where does it come off? Comes off the bottom there, doesn't it? Is come off the bottom, yeah. So what does the bottom of the separator look like?

说话人 2 01:24:02

Is that that the green pipe that, yeah, it's the green pipe that comes off the bottom of the separator. Where does. Can we look at the back of the skid so we can see the separator discharge? OK. on the point.

说话人 1 01:24:21

咱们从后头看, 从翘块的背后看一下, 转, 对, 然后看一下他分离器底下是怎么出来的。

说话人 2 01:24:50

So where to that? The line off the bottom of the separator goes through to the front of the skid. Where's, where does the outlet from the separator, the process outlet?

说话人 1 01:25:05

咱们, 他的工艺的出口在哪块啊? 这个。哪个? 这个。

说话人 3 01:25:13

就是我我点的这个。

说话人 1 01:25:16

the purple one, She and shuangshuang is pointing at here. This is the process. Oh, that.

说话人 2 01:25:30

Definitely this one. Definitely. Can we see where that come? Cuz there's a t.

说话人 1 01:25:41

从分离器的哪块出来的? 这是。

说话人 3 01:25:43

从分离器的底部，这不是分离器吗？然后这是分离器的底下那个管路，从底下管路出来之后，减压阀，控制器，是电磁阀，开关阀。

说话人 2 01:26:00

So that line just goes into the back of the pipe. Yeah.

说话人 1 01:26:04

你能把交汇点，OK，就这样。

说话人 3 01:26:08

交汇点在这，要从这。就是那个主管从那，对。

说话人 2 01:26:14

so if we put the valve just downstream of that tee, that minimizes the dead leg.

说话人 1 01:26:26

OK，他如果把阀，放在这个梯形管的下游，是不是就把相当于就把所有的死区就给。减到最小。

说话人 3 01:26:43

啊啊，他是把这个梯形管放到这块，应该是吧？

说话人 1 01:26:49

不是，我们把把阀放在这个梯形管的下游口。

说话人 3 01:26:55

啊啊，但是这就不好操作了嘛，当时不是说把它引到端面来。

说话人 1 01:27:02

这个阀就改成一个气动的，引到一个球阀上，球阀给气，我们操作那个球阀就完事了。

说话人 3 01:27:10

这我明白他的意思，那会意思是说，就这个出口。出口到排放阀的管路应尽量短。然后如果不便操作的话，就变成一个气动阀。

说话人 1 01:27:22

嗯，对。

说话人 1 01:27:26

Yeah, so we, the the we, our understanding is we put the valve, the dream valve on the downstream of the tea and then making it pneumatic actual valve. And.

说话人 2 01:27:46

And if, yeah, if the guys can't reach it easily from the end of the skid, yeah, they won't help to reach one from the end of the skid. But yeah, pneumatic valve is, but.

说话人 1 01:27:58

yeah, pen pneumatic valve, yeah, 对，它就是这个意思，变成气动阀。

说话人 3 01:28:08

这是这个，但是不好几个放，排净阀嘛，都要变成，都要变成气动阀。

说话人 1 01:28:16

yeah, there. Several similar situation for the June wall. Do we all do this similar design, minimizing the lens and making it the pneumatic?

说话人 1 01:28:32

包。

说话人 2 01:28:34

点。就。

说话人 1 01:28:35

对。

说话人 3 01:28:35

不是，李工，就是还是说，就说，比方说，你看我原来这个阀，就这个排气阀，是在这个位置的。就是可能，就是需要人进去接一下，这种如果有空间的话，是不是就可以不变了？就是实在是没有空间够得上这个，然后再变气动阀。

说话人 1 01:28:53

我听他这几次讨论，我感觉他就不想让人进去了。就不要给空间了，就直接变气动的，从外头操作，里头不留空间。然后拽，也跟咱们的想法是。你说对。就是特别冲撞的一点，但他就是说就不让人进了，就进不去的，这意思。

说话人 3 01:29:17

这个其实这还是可以进人的，如果把这个阀留在这个的跟阀的底部，进去接一下也行。但是你说他，他他，既然他这个设计的想法是。

说话人 1 01:29:30

So, Shuang was thinking that you can see there are some space that for a person to come in. Okay, okay, okay. And so if we're putting the wall on the downstream of here, what she is pointing that someone could go in and operate. Do you still thinking that this kind of. Spacing in space for person to come in, design is acceptable or we just no longer thinking about let some people go in and operate this kind of thing. Let's put the valves in the best place and then see how far they have to reach in.

说话人 2 01:30:15

If they're reaching in 3,400 mill, then potentially they can operate the valve. Just reaching into the skid. They're not actually going into the skid. They're just reaching in. But you've probably got about a 400 mill reach into the skid. Yeah. Okay, so there's two step there. If we end up. Actuating more valve than we have manual for the operator, it probably makes it easier if we just actuated all. There's only six valves, isn't there? Yeah, six walls. Yeah. So you had six. Pilot valve on the edge of the skid because you can't access many of them. That makes more sense if you can access, if it's only one actuated valve, maybe two, maybe that makes sense to operate it remotely. It's trying to make it the same operation for the operator. So he either does the majority or all of them. Manually or he does all of them with a remote actuation.

说话人 1 01:31:33

OK, understand。好，范工，他的想法是这样的，如果他就是三四百毫米的话，他觉得伸手能够到，你放在最合适的位置，他在三四百毫米的状态下伸手能够到，他确实可以手动开关阀。同时还有一个原则是他希望最好就是你，比如说你大部分都是远程控制的气动阀，这样的话给这个操作工一定的一致性，比如说一，他一共6个阀，一两个是手动开关，大部分都是气动的，或者就是大部分都是手

动的，一两个气动的，这种的状态，最好就是比如说要气动都气动这种状态。就这样的话，不会对这个他们的那个。操作工造成困扰，可能他们也就是，就乱了。说这个手动，那个气动，它就比较乱。

说话人 3 01:32:40

行，我明白了。还有这是排进管，那个就像那个冲洗管路呢？这个 CIP 管路。我就是引到这个跟前来，这种它算不算死区呢？也算死区吗？你看比如说是从那么上来，过去之后回头，这块也是存液的，这都要换成气动阀吗？

说话人 1 01:33:02

在，刚才那个是什么？CIP 管路啊。刚才那个，这是 CIP，刚才那个呢？

说话人 3 01:33:09

这个也是，就是两个 BT、BT01和 BT 04的 CIP 管。

说话人 1 01:33:13

他刚才说一共有6个，是你。

说话人 3 01:33:16

6个，那。

说话人 1 01:33:18

你你想想一共是几个？

说话人 3 01:33:20

我想想啊，1234，这不4个吗？还有吗？

说话人 1 01:33:24

就是总共你觉得需要改成气动的，你。

说话人 3 01:33:28

不是，四、5个、5个，我，哎，6个吗？这，就是12，然后刚才那两个，然后还有一个分离器应该有一个，他说有6个，是吗？

说话人 1 01:33:39

就包括那个排净，包括这个 CIP，都包上，一共有几个？

说话人 3 01:33:45

我现在数的是5个呀。5个是吧？

说话人 1 01:33:48

他说6个，我看看啊。他就随口一说，不一定是对。你你你确，你你最了解。

说话人 3 01:33:54

我现在我觉得是5个。

说话人 1 01:33:57

就是 CIP 跟排净都算上，是吧？

说话人 3 01:34:00

他他那个BT04的出口如果也算一个也算一个排净的话，那就是6个，嗯。

说话人 1 01:34:05

嗯，那就他就说，那他不就是说这意思，6个，要不然都变成。

说话人 3 01:34:12

BT04已经是个已经是个那个，

说话人 1 01:34:14

气动阀了，OK，

说话人 3 01:34:16

对对。

说话人 1 01:34:16

那要不然都变成气动的，然后要不然。

说话人 3 01:34:19

都变成气动的。

说话人 1 01:34:21

他觉得。

说话人 3 01:34:21

都变成气动的现在。不是，现在主要是都变成气动的话，气动阀得需要很多的安装空间。你看他这一个气动阀，上次跟高工一块看的，他这一个气动阀是一个空间，他还有一些个定位器，还有这，占的空间挺大的。他就是相当于你就在这里边再增加几个气动阀，我觉得这个空间会很费劲。

说话人 1 01:34:48

什么？他他的意思就是说变成气动以后，他有一个球阀来控制他，球阀可以放在这个好好弄的腔那块，是这意思。你，他不一定是启动阀，就是一个。驱动器相当于。

说话人 3 01:35:02

驱动器那也不，驱动器就除了电动阀就是气动阀。我就觉得气动阀反正是不太合适，因为气动阀现在没有量嘞。就是你看，他说如果增加再增加五六个，我觉得是没有地装的。

说话人 1 01:35:19

那不就是增加5个吗？

说话人 3 01:35:21

5个对，5个。

说话人 1 01:35:24

那你，那要不然试一下，你看看就是把那个，相当于他的原则是把阀门放在最合适的位置上。再弄成远程驱动的状态，试一下，看看是不是空间不一定够。

说话人 3 01:35:43

对，空间现在肯，我觉得肯定是不够的，因为底下这个地。那你排一下给他看一下。

说话人 1 01:35:50

行。

说话人 3 01:35:50

我排一下吧，就按他说的那种原则我排一下啊。

说话人 1 01:35:53

行，耶。

说话人 3 01:35:54

伸手够着。嗯，OK。行。

说话人 1 01:35:56

So, hey, Keith. Hi, Keith. Yeah, we just counted. Yeah, it's about six pneumatic valve and we already got one. The BTO4 is already pneumatic. And for the other five, and according to Gao Yu and Shuang Shuang, it may consume a li, a large amount of space for the pneumatic valve. So, shuangshuang was worrying about if we got sufficient space or not, but she will try that to arrange the space and to see if we can have all the 5 pneumatic valve together with the BTO4.

说话人 2 01:36:46

Okay, let me just share my screen again. The. If we go for a standard half inch ball valve with something like an air talk actuator on it, I think they take a much smaller space than the swage lock valves. Oh, that so, okay, then go ahead. Sorry. It might be a case that if you've got a 10 mill pipe. But for things like the CRP where you want the volume through it, it might be a half inch pipe. A 15 mill pipe is easier with a standard valve, standard haven't ball valve and an actuator that would be a smaller footprint than.

说话人 2 01:37:39

嗯，a switch lock。

说话人 1 01:37:42

OK，高工在吗？高工我看在，他这个东西，他现在正在共享这个，是一个球阀，加上一个气动的这个驱动器的一个阀，他说这个东西比斯宾奥克阀要小很多，他比较适应1/2英，半英寸这种状态的的管径大小。说我们建议我们可以用这个东西，不用斯宾诺克那个比较大的没有必要。

说话人 6 01:38:22

哦，行，可以。那把这个厂家可以发过来看看资料参数什么，那个厂家资料样本。什么的，我们可以寻一下，看看我们这边国内有没有能买到这个东西。

说话人 1 01:38:38

行。

说话人 1 01:38:39

So, do you have any suggested vendors for this kind of the ball? We sent you the Haban data sheet for some valves.

说话人 1 01:39:03

他之前给咱们发了一些阀的数据表。

说话人 2 01:39:15

数据表。

说话人 6 01:39:32

哦这个是汉博尼的，我知道，之前他给过这个。

说话人 1 01:39:36

对，他应该就说用这个的。 Oh, this the.

说话人 6 01:39:51

嗯，对，这个是我们被选的厂家之一，这个当时。

说话人 2 01:39:56

Yeah, and it's our candidates vendors as well. And this one fits well. Yeah, and they do their own actuators as well. So it rather than an air talk, it might be just as easy to use a simple habanero actuator.

说话人 1 01:40:18

yeah, got it. 他就是说比用起 airtalk，还是用这种汉博尼的启动可能更合适一点啊。

说话人 6 01:40:31

啊，好的，那个汉博尼我看跟刚才您发的图片里好像不是一个厂家。

说话人 1 01:40:36

对，他他图片里头，他就是给个例子，他就。啊，明白。随便从谷歌上搜了一下啊。嗯嗯，好的，明白。然后我就顺手赶快截图，怕有需要啊。

说话人 6 01:40:49

汉博尼他之前给过，知道这事。嗯，行。好。

说话人 1 01:40:52

Yeah, got it. Thanks for the information.mission。

说话人 2 01:40:55

yeah, now that' s fine。

说话人 1 01:40:56

OK，那行，范工，我们就按这个，到时候高工这边确定一下尺寸大小，给到你数据，我们按这个来，这个思路先排一排，看看能不能排开。

说话人 2 01:41:12

Oh, hold on a bit.Okay.Oh, Peter, I was just looking.They, it looks like they will do a class 9038 valve anyway.So yeah, you could probably stick with your 10 mill three eights and put one of those valves on there.

说话人 1 01:41:33

Okay, so it's covers a larger range and we can just.keep our design。OK，got it。那应该是对，我们现在对应的大小的，应该都有。就管径大小的，应该是有这种型号的，他刚才看了一眼。行，那我们继续往下快速的过一下。其他就是同步信息的事嘛，是吧？

说话人 2 01:42:12

So it's, we have three piece valves on site.They do class 900 and smaller sizes.So yeah, that could well be flange connection if you need a flange. Okay.

说话人 1 01:42:31

库，它也是可以访问链接的。因为他们在他们现用的就有类似的型号，应该可以找到。

说话人 1 01:42:45

Okay, and there is like two or three more questions, but it's just some confirmation. And this one is about the platform and the. Charging the PLA, the ladder or the platform for charging the going high. This one. We just want to confirm this one. The design of the platform will be in charge of the sandwiches. That is that right? And procure by. By subsidy and you guys were gonna fabricate. Okay, is that right? We just want to confirm the design will be by your side.

说话人 2 01:43:58

We can give you dimensions, and then mechanically build.

说话人 1 01:44:06

would that be something you guys could do? 赵工，赵工在吗？

说话人 7 01:44:17

在呢在呢。

说话人 1 01:44:18

到底要确认什么呀？他们给 dimension，那个维度，咱们造，是吗？还是这些需要确认什么呀？这句话。当时要问的，我看，还是这个不需要跟他说，我不知道啊。

说话人 7 01:44:36

上次开会的时候，这不是，这不，不是我写的，我不太知道啊。上次开会的时候，你不是说过了吗？这个平台和那个梯子。是 UK 那面设计，然后咱们这面，咱们按照他那个图加工。

说话人 1 01:44:56

嗯，对，这块，那是高主管您写的吗？是奔博士写的。

说话人 3 01:45:05

这是，不，这是我写的，那会写了，要确认一下，我就说再明确一下。是在。

说话人 1 01:45:13

张工。那个0922是你写的，是吧？

说话人 3 01:45:16

对，0921，那个0921也是我写的，对。

说话人 1 01:45:20

就是0915那个是奔博士写的，是吧？嗯对。我就怕理解的有偏差，你们看看，就是最终我理解赵工就是3D位置设计，咱这边按。对。平台有3D位置设计，是吧？咱这边加工就行了，是吧？就确认一下，避免。消息对。对，这个现在已经确认了，是这样的，他会给各个尺寸啥的。

说话人 7 01:45:51

我想问一下这个大概的时间。会不会，因为为，他这个事啊，他这个梯子平台是想和那些主设备。一个流程出去，就是找一个供应商，这样的话一起认证他方便，要是找两个供应商是比较麻烦的。OK，就。

说话人 5 01:46:08

那个 pita 那个 t，pita t t 选说是，那个 key 去 design 这个东西，之后我们去。外部。对。对，不是我们 design，机器是信息会确认。

说话人 1 01:46:21

对，这个他刚才说了，是他们 design，他给那个尺寸。OK。对呀。

说话人 5 01:46:27

对。那就去那那个贴，他有的贴全，我今天发给那个发工了，发工看看吧。

说话人 1 01:46:33

那个图。

说话人 3 01:46:34

林博士，那个不是他的，他不是那个，那不是他的那个那啥，他那个图只是原来那个 case 他想要一些个尺寸，去设计这个梯子，他居然想要咱们用，咱们来提供，后来他就是把我给他那个 STP 格式，他自个转了，转出来之后他就可以把那些尺寸自己标出来了。就是他现在应该是不需要咱们提供尺寸了，因为他自己已经在他那个图上标出来了啊。

说话人 5 01:46:57

ok。

说话人 1 01:46:59

Yeah, high case, we just discuss and yeah, we just made sure that you will design the platform and we will fabricate. And so it's confirm and or wait.

说话人 2 01:47:18

That doesn't work well for us. If we get somebody, we would normally give somebody the dimensions of the platform, okay? They would go away and design it to meet the standards and then fabricate, okay, they make the money when they fabricate, okay? They don't make the money. They design.

说话人 5 01:47:39

Okay, so to get somebody do us a design for somebody else to fabricate doesn't work well. Okay, so you will give us the dimension. Sorry, Keith, it's Ben here. How Ben? Sorry, Keith. Yeah, I think last week we discuss about the staircase. So the staircase, you will do the design, right? Or we will do it.

说话人 2 01:48:01

We, I think we, if we do it, yeah, we would need to place, we would need to do all of it. So we would go out to one of us metal fabricators, and I would build a set of steps to our requirements. Okay, so they take our dimensions and then they make it. To meet the British standards, they wouldn't do us a drawing that you could then fabricate because they would, they make their money on the fabrication, so they wouldn't be interested in just doing a drawing. That's not what they do. Yeah, that's fine. So we can do all of it. Yeah. We give you dimensions and you do the, in effect, you do the design and build based on our dimensions.

说话人 5 01:48:50

yeah, it's fine, so we can do it, yeah, OK。

说话人 1 01:48:51

yeah, OK。就是刚才又确认了一下，是他们给各个的尺寸，咱们设计跟加工，还是得咱们设计，因为他的意思是说他们要找一个第三方设计的话，没人会接，因为必须得把设计跟加工同时给，他们在加工里头是挣钱的，他如果是光设计的话，他们不挣钱，就没人会做这个设计，所以最终。按这个想法来讲，还是得咱们来设计，他们给尺寸，咱们尺寸跟要求，咱们来根据这个尺寸跟要求来设计，设计完以后咱们自己制造。

说话人 7 01:49:33

这个梯子平台大概会有多大呀？

说话人 1 01:49:41

so, hike is, is there like。

说话人 2 01:49:45

A gut feeling of the.How large is the platform?

说话人 2 01:49:56

I'm hoping it'll be less than 1.4 meters high because that was one of the, once it goes above that, it becomes more stringent standards.And so it's more of a couple of steps up, take you up maybe a meter, and the guys would need to manipulate a bag. So.I'll draw it out, maybe 8,800 square with some steps coming off it.Okay, got it.All he's gonna do is go out some steps, hang a bag on a hook, manipulate, connect up the flange, manipulate the bag to get the catalyst to flow down into the reactor and then refit the flange. So it's not, he hasn't got much to do.Yeah, when he's up there, he just needs a little hop up to get up to that level. Yeah.

说话人 1 01:50:52

对，他应该就一米多长，一米多高，不要到一米四，他八百的的一个正方形吧，在最上头。他其实他意思说做的事不多，你就把东西给挂上，带一个袋子上去，把东西给挂上，往里头倒料啥的，所以就应该这样就差不太多了。

说话人 7 01:51:26

这是。我是感觉他这个东西应该不是。

说话人 1 01:51:28

基础的那种感觉啊。

说话人 7 01:51:30

不是太复杂的東西。对。他是在英国附近，他不好买这个东西吗？是希望咱们给提供吗？还是。

说话人 1 01:51:38

就是咱们。考虑的呢？对于咱们来说是，这个东西是这个由英国这边买更方便，还是以，咱们出钱由英国这方面那块买更方便？还是说咱们买完送过去更方便？

说话人 7 01:51:56

我觉得还是他那买更方便呀。因为咱们买，你还得再认证啊。

说话人 1 01:52:03

OK。

说话人 2 01:52:04

So, no, wait, it's something, yeah, I kind of envisage something like this made out of stainless steel rather than aluminium. So it's, it doesn't give issues with sparking, wheel it in. There might be a little shelf on here if they need a shelf to hold something, maybe a little hook to hang the bag on so they can hang the bag. Go up, pick the bag up and something like that. That's what I kind of envisage.

说话人 1 01:52:38

Yeah, so I thought there's already like a price and vendor there in UK. So we're thinking that if we're buying it from the UK, is that more convenient? Cuz Zi Liang was worrying about the CE Mark. And if we're going to fabricate design, fabricate it ourself. Then it will go with the main equipment. And this one will add more work on the sea marking. But if we're buying it directly from Own Local UK, then it get rid, we get rid of a lot of.

说话人 2 01:53:19

See, marking issue. Yeah, we can do. The only thing we probably wouldn't buy one off the shelf like this because it's aluminium. Oh, okay. So we would need it made out of stainless. So the price goes up quite a bit. But. Yeah, we can get one made in the UK. Yeah, if that better for you guys, it probably ends up being a little bit more expensive than if your guys make it. But by the time you certify it, it might work out cheaper. Yeah, I think that more made in.

说话人 1 01:54:01

That a little bit more cost would worth it. Let me double check with Zhang.

说话人 1 01:54:08

他的意思是说他们那块可以买，他这个东西686是铝的，他需要的是一个钢的，钢的肯定更贵，比这个686肯定贵很多，他觉得说如果我们自己造的话会便宜，他他们那块买是方便的，会贵，这样如果是CE认证这个事，咱们这块更方便的话，那他们那块买没有问题，就让咱们来承担一下这个更贵的，咱看评估一下是否能承担这个更贵的这个。对，比自己造更贵的一个情况。

说话人 7 01:54:48

他这686是英镑啊。

说话人 1 01:54:51

对，英镑，是铝的还是？钢的还会更贵。

说话人 7 01:55:01

他英镑和人民币差的。

说话人 1 01:55:04

1:10或者1:9吧，你就按1:10算。

说话人 7 01:55:08

6800。不锈钢的他说能更贵一点。对。6800，再贵一倍，也就是6万多呗。

说话人 1 01:55:21

现在是1:9，嗯。

说话人 7 01:55:23

不是，我的意思是，你现在按铝的话，这不是6800吗？要按6800来算，如果不锈钢的话再贵10倍，那才6万8呗。

说话人 1 01:55:32

对，10倍是6万8，但应该到不了10倍吧。

说话人 7 01:55:35

我感觉到不了10倍吧。有5倍应该差不多了，3万多块钱。我觉得你这个一个东西的认证。和那啥买这些材料。

说话人 5 01:55:48

不是那个小龚啊，我在你余额里面东西跟主设备没有很大关系，就分开弄的话，就他们买吧。

说话人 7 01:55:57

是，我也是这样想，他和主设备他关系不是很大，他可能带来额外的东西。

说话人 5 01:56:02

他们就直接直接直接属地去配吧。

说话人 1 01:56:07

Hold on, Xiao.Yeah, so after discussion, we think it make more sense to buy from UK and, and so, and then it, it cuz it not that related to the main equipment. So the CMR will cost more maybe so.

说话人 2 01:56:29

Now we do that, we'll, okay, we'll supply the steps. That's fine. Okay. Yeah, that would be.

说话人 1 01:56:35

great, thanks。

说话人 2 01:56:37

my problem。

说话人 1 01:56:41

诶，行，那这块就确定了，从属地去搞了，好吧？我们我周三明天的时候在总结会上也说一下这个事，好吧？

说话人 1 01:56:58

Okay, yeah, and thanks.This one is solved.And this one is f by du Zhi Desmond.And she said, he said he has already provided the minimum size of the Join Junction box with email. And you can.Check it out and to see if there are suitable size and you can we can purchase and we can.Discuss deeper about how to, if you want, we want to split that one more.Like we were talking about pulling out one skill.And so if we're splitting the control panels, that making less where disconnection and installing and that maybe make more sense.

说话人 2 01:57:57

But the Desmond hasn't designed that part, but we can explore more on this.Yeah, I think most of them go into that, the panel that would fit on the reactor subassembly, so that makes it easier.I was talking to the guys on Friday about the panel, so hopefully I can feedback.Any progress Friday or Monday? Okay, that would.

说话人 1 01:58:22

be great, thanks。行，那个孟工，他刚才说我刚才说在这个，如果这个考虑那个分翘往外拉的这种情况，咱们再设计设计，再有个更深的、深入的谈论。然后他说他现在已经去寻了，周五或者周一给咱们一下更新。Okay, okay.Oh, no, maybe today.

说话人 1 01:58:49

Yeah, the last one is about the procurement raising by Zi Liang.And so we just want to make sure the main equipment drawing, we're planning to do the.The procuresment in these days.And so that it can, we can meet the timeline.

说话人 1 01:59:16

Current timeline.So we just want to verifying the equipment drawing and which the latest mean equipment drawing is send issued on by email on August 18th. And is there any.Thing comments or 0 objections from your side, we can double check and verify that.

说话人 1 01:59:43

And the current plan is that the, this, oh, the fir, the second week of the October, we may finalize all of the.Main equipment drawings and so that we can use this one to code and to talk with the main equipment vendors and do the procurement.I haven't been through them yet, Peter, to, okay, to be honest, I was waiting for the model.

说话人 2 02:00:17

Cuz I think the majority of the comments would be around nozzle orientation, okay?And until we get the model, we don't know what orientation we need.So that, that was, that's the reason why I've not looked at them is.Until we know they fit and we can say, yeah, I've checked all the nozzles, they're all in the right direction, we know it fits, it's hard to say they're good to go.Okay, got it.So we need to finalize the model first and then do this one.Yeah, or at least get enough confidence that the nozzles are in the right place.And okay, the things are the HTF, we, the pipe runs are good.Okay, you know, we don't necessarily have to do the rest, but we know they need to go in that direction and it's good. Okay, got it.

说话人 1 02:01:14

赵工，哎，赵工听得到吗？

说话人 7 02:01:18

哎，我听得到。

说话人 1 02:01:19

OK，他其实主设备这块他没太仔细看呢，因为他觉得管嘴的朝向啥的，这些的是一个重大的情况，所以得，他认为先得把3D模型里头主设备相关的部分都给确认好，他的 HTF 的怎么接？各个相关的管嘴的朝向，管怎么接？这些都确认好，我们才能定下来这个主设备图纸，才去采购。所以就是现在瓶颈是3D模型这块，不是图纸这块。就是图纸不用全定下来，但是关于主设备的跟一些管道连接都得定下来。

说话人 7 02:02:09

嗯，他还是想等3D模型审好了，确定好了之后再研究图纸。

说话人 1 02:02:15

对对，是的。

说话人 7 02:02:18

行，那正好，这不也是他们下个月去完。去完 UK 之后。模型定好了之后，咱们不，领导不是让重新再排计划吗？对。那就只能那会的时候根据实际情况再排了。

说话人 1 02:02:33

行，可以，OK。

说话人 1 02:02:35

So we understand. So let us finalize.

说话人 5 02:02:40

皮特说。皮特。公公凯总跟我说，Keith 可能11号那个星期有问题，他可能我们后，又往后移那个去英国的事。

说话人 1 02:02:54

往后移呀，啊。

说话人 5 02:02:56

对，往后移三周，所以刚刚在公司的那个东西后，你在这几周里面，好了，就这边弄吧，好了，你等不了那个去那个现场的，可以看看吧。

说话人 1 02:03:07

对，就是咱们也没必要就完全等这个，就是去现场，以后再把这些事给确定，它实际上的 bottleneck 就是就3D模型，主设备相关的这些部分，咱们不用等其他的東西，这个3D模型相关的部分确定了，咱们就可以往下走，好吧？

说话人 5 02:03:28

赵工。对，因为高工。凯越跟我说的，就是那个他拿信息好像好像好像是有事，我一会就跟他确认一下吧。

说话人 1 02:03:36

行，OK。

说话人 5 02:03:37

所以这个去英国来一个，又移到11月11月初了。

说话人 1 02:03:41

OK，好的。

说话人 1 02:03:43

Okay, issue, you know, yeah, so I just talk to Ben and Zi Yang and yeah, we understand the, we need to finalize the main equipment related model part and also the connection of the hdf part. And then we. We can finalize the drawing. And so we will first solve this part and then. Do the, do the finalize join and then do the procurement submission process.

说话人 1 02:04:17

Yeah, excellent. Thanks, Peter. Okay, thank you. And that's all from us. Excellent stuff. That's brilliant. I'll, I think we're meeting again tomorrow, aren't we? Tomorrow morning. Yeah, tomorrow. Excellent stuff. I'll speak to you in the morning then. Okay. Sorry, your evening. Yeah, how are you doing? Okay, thank you. Thank you. Soon. Yeah, thank you. Bye bye. Thanks.